

Evaluation of the E-contracting reference architecture

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**Evaluation of the E-contracting
reference architecture**

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Evaluation of the E-contracting Reference Architecture

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Abstract

Reference architectures provide the major guidelines for the structure and functionalities of a class of information systems. Because of their fundamental role, reference architectures must offer a high degree of excellence. In our previous work, we have defined an initial version of an e-contracting reference architecture. In order to determine its degree of excellence, we performed an evaluation of the initial design. In this report, we present the evaluation process and the main results from it. We discuss the improvements that were introduced to the initial design of the architecture due to the evaluation process.

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1. Introduction

Contracts between value-exchanging sides have been adopted to specify the exchanged values and the rights and obligations of the participants. Business-to-business e-contracting uses information technology for improving the efficiency and effectiveness of contracting processes of companies.

As in most modern software development projects, the development and implementation of a highly automated e-contracting system must be preceded by the effort- and time-consuming "system analysis" and "system design" phases [1]. The existence of a specification of the requirements on an e-contracting system and of a reference architecture to serve as a guideline in the design of concrete architectures will significantly facilitate the software development process. An E-contracting Reference Architecture (ERA) will introduce a number of benefits to software developers and business professionals. ERA will allow faster development of e-contracting systems. Furthermore, it will introduce standardized view on e-contracting systems, allowing modular system development and facilitating interoperability with other information systems. ERA will lead to improved understanding of e-contracting systems.

In [2], we have presented our initial design of an E-contracting Reference Architecture (ERA). Being a reference architecture, ERA should offer a high degree of excellence. For this reason, after elaborating the initial version of ERA, we have evaluated ERA for its suitability and excellence as a reference architecture. The goal of this evaluation was to discover if there are points for improvement of ERA. Results from the evaluation would allow us to achieve higher degree of excellence in the design of ERA. In this report, we present the process of evaluation of ERA. We discuss the main improvements of ERA that resulted from its evaluation. The final version of ERA that resulted from the evaluation process is provided in [3]. This report has to be read in combination with [2] and [3]. It serves as an additional explanation for the transformations that took place in the initial version of ERA.

The report is structured as follows. In Section 2, we present the method that we used for the evaluation of ERA. Section 3 presents the evaluation process and the results from it. The report ends with conclusions.

2. Evaluation method

Literature provides a set of methods that can be used to evaluate architectural qualities. Brief summaries and comparison of the most popular methods can be found in [4] and [5]. These methods originate from the software engineering domain and are targeted for the evaluation of concrete software architectures. Next, we explain the method that we used for the evaluation of ERA.

In the initial version of ERA [2], we have identified the following system and architecture non-functional requirements (see Figure 1):

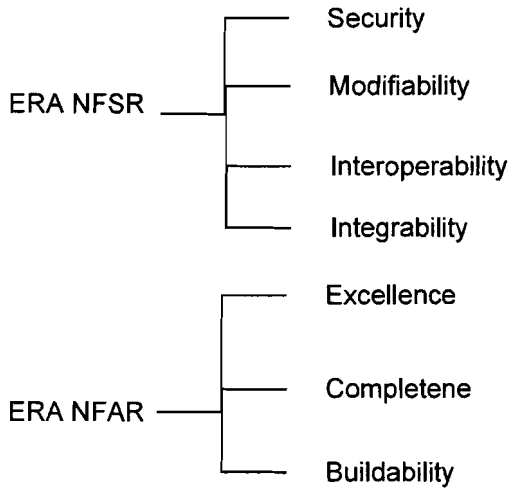


Figure 1 - Non Functional System and Architecture Requirements (NFSR and NFAR)

Based on the comparisons presented in [5], we concluded that from the existing methods for architecture evaluation the Architecture Tradeoff Analysis Method is most suitable for the evaluation of ERA. The Architecture Tradeoff Analysis Method (ATAM) [6] supports evaluation of a wide set of quality attributes. It is considered to be very suitable for evaluation of modifiability, integrability, and interoperability attributes [5]. ATAM concentrates on the relation between the architectural approaches used in an architecture and the desired quality attributes, and investigates potential sensitive or conflicting points (called “trade-off points”). Based on the analysis, conclusions are drawn on the potentially problematic and positive aspects of the architecture (risks and non-risks).

Our initial goal was to use ATAM for the evaluation of ERA for the modifiability, integrability, interoperability, and security quality attributes. Furthermore, we wanted to investigate the applicability of ATAM for evaluation of the other quality attributes. Next, we provide a summary of our results from the evaluation of ERA via ATAM and the additional activities that we performed for the evaluation of ERA. As ATAM is intended for evaluation of concrete architectures and ERA is a reference architecture, we had to adapt ATAM in a number of points for the evaluation of ERA.

3. Evaluation of ERA

In this section, we explain the steps that we followed for evaluation of ERA. In ATAM, the initial design of the architecture is followed by an investigation and analysis phase. In the next phase, the results from this analysis phase are tested. In this section, first we discuss these two phases in the evaluation of ERA. Next, we discuss the additional steps that we took for evaluation of the quality attributes that could not be evaluated via ATAM.

3.1. Investigation and analysis of ERA via ATAM

As stated in ATAM [6], the first group of steps that have to be undertaken are analysis and investigation of the architecture. In this phase, we organized three small meetings within our research group involving 10 software architects/designers in the first meeting and 3 in the second and third. In the first meeting, we conducted a general, unstructured discussion on ERA. After this meeting, we elaborated a draft list of the architectural approaches (styles, patterns, and other reference architectures) used in ERA. In the second meeting, we discussed the quality attributes and improved the utility tree presented in Section 2. As a result, a new quality tree was defined (see Figure 2 and Figure 3). We decomposed the attributes that we planned to evaluate through ATAM to concrete scenarios. Concrete scenarios were ranked for their importance and for complexity of addressing them in ERA. Based on the new utility tree, we discussed also improvements of the list of architectural approaches.

As can be seen from Figure 2 and Figure 3, the amount of quality attributes has increased and our understanding for the required quality attributes in ERA has improved. A major problem that we faced at this stage of application of ATAM is the decomposition of the quality attributes and the elaboration of concrete scenarios for these attributes (as it is required in ATAM). We realized that definition of scenarios for a concrete architecture is natural and different scenarios can be easily generated (use-case, growth, and exploratory scenarios). However, in the case of a reference architecture, the scenarios that we defined were very general. The reason for this is that a reference architecture has no context which can provide concreteness to the scenarios. Due to the general nature of scenarios, prioritizing them was also hard. In a general case, most scenarios were of equal importance for the architecture.

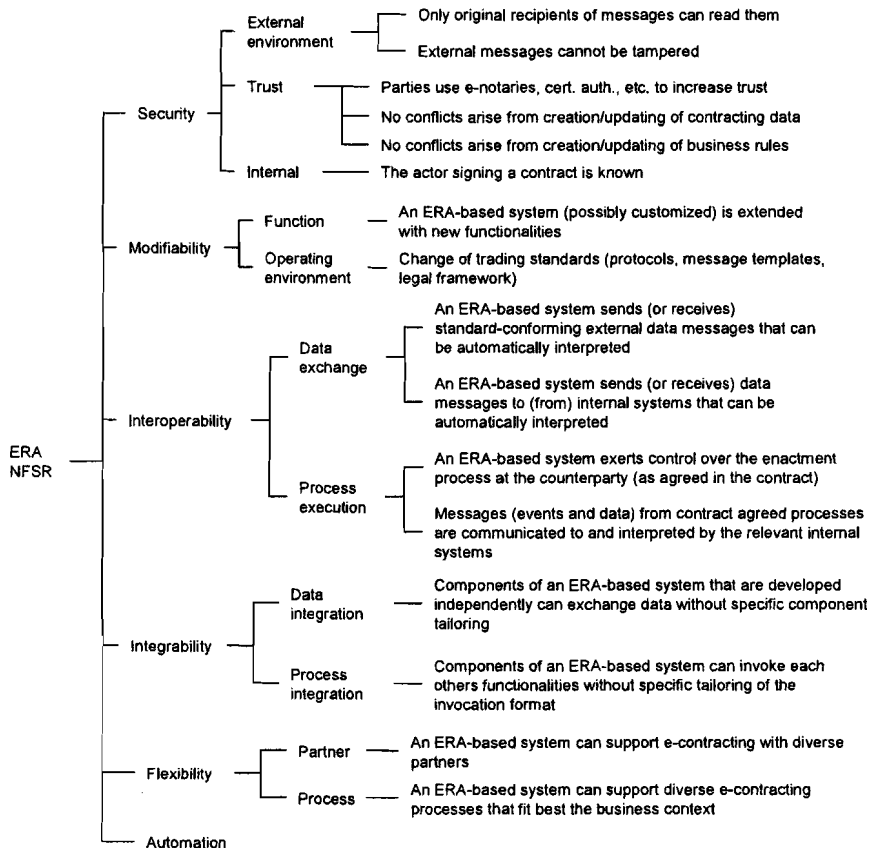


Figure 2 - Final utility tree - Non Functional System Requirement (NFSR)

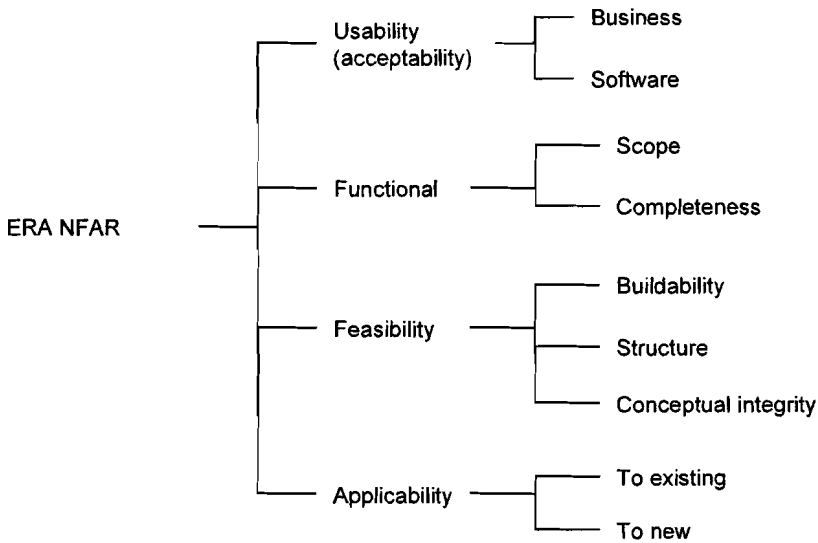


Figure 3 - Final utility tree - Non Functional Architectural Requirement (NFAR)

Next, we used the list of architectural approaches applied in ERA to analyze their suitability for the quality attributes already defined. We used the template forms suggested in ATAM for this purpose. In Appendix A, we list the forms that resulted from this step.

During the analysis process, we found a number of weaknesses and points of improvement in ERA. To address these problems a number of new architectural styles were introduced. The lack of a consistent strategy for exchange of data among components and for invocation of components lead to the introduction of the Data Indirection style, Abstract Data Repository style, Façade pattern, and the Broker pattern. We discovered also that a new component (i.e., Internal Mapper component) had to be added to the architecture to address the requirement for process and data interoperability with internal systems. Finally, we defined the following list of architectural approaches used in ERA:

Architectural approaches at the first level of ERA

- Layering style [7], [8].
- Part-whole decomposition style [7].
- Facade pattern [9].

Architectural approaches at the second level of ERA

- WFCM standard architecture [10] and RAWFMS (Grefen and Remmerts de Vries, 1998).

Related components: *Contracting Manager*

- Batch sequential pattern [7].
Related components: *Secure Messenger*.
- Data indirection style [8].
Related components: *Selection rules, Negotiation rules, Enactment rules, Management data, Application data, Process data, Process, rule, and ontology definitions*.
- Abstract data repository style [8].
Related components: *Invalid messages, Management data (PS), Evaluations, Management data (C), MC/SA, Management data (E), Management data (SM) Rejected updates*.
- Broker pattern [11].
Related components: *Internal broker*.
- Mapper pattern [12].
Related components: *External Mapper, Internal Mapper, Enactment Mapper*.
- Security dedicated components - *Cryptographer, Signature manager, Signer, Communication Monitor, Updater, Contract distributor (MC/SA)*.

Based on the final list of architectural approaches used in ERA, we have identified the following relations between architectural decisions and quality attributes:

The **modifiability** attribute is addressed by the usage of the Layering, Part-Whole decomposition, Abstract Data Repository styles and the Façade, Broker, Batch-sequential, and Mapper patterns.

- The Layering style facilitates achieving higher degree of modifiability. Changes in functionalities influence only “higher” layers. A company may initially implement few components from the application layer and add additional components in the future. This will have impact only on the Contracting Manager component.
- The Part-Whole decomposition style promotes achieving *separation of concerns* among components, which leads to increased modifiability of a system.
- As explained, in ERA, components from different layers exchange data through databases that implement the Abstract Data Repository style. This style is used to achieve *indirection* in data exchange.
- We expect that a component from the first level of decomposition of ERA will be developed by a single company. Consequently, data integrity between data providers and data consumers becomes of lower importance. For this reason, for data exchange within components, we opted for the Data Indirection style, which also supports high degree of modifiability (easy additions/removal of data consumers/providers) but does not abstract from the format of the stored data. If differences in the data format are to be expected, the Abstract Data Repository can be used.
- The Façade pattern is used to provide *encapsulation* of “lower-layer” components from “higher-layer” components. Using facades allows a “higher-layer” component to be unaware of the concrete decomposition and implementation details of lower-layer components.
- The use of the Broker pattern facilitates achieving *indirection in communications* between internal systems and an ERA-based system. Internal

systems can be relocated, added, etc. without affecting ERA but only its Internal Broker.

- The use of the Batch-Sequential pattern allows easy modification to be applied on the Secure Messenger. As this component may be shared with other systems that require external communications, it can be easily adapted to fit their requirements.
- The Mapper pattern is used to define components that “absorb” changes in the external environment with respect to data and process semantics (see the Internal Mapper, External Mapper, and Enactment Mapper). This pattern reduces the amount of modifications required by an ERA-based system when modifications in external data occur.

The Mapper pattern is used to address the **interoperability** attribute as well. Mapping components support the usage of different data and process specifications in an e-contracting system and external/internal systems, which leads to increased interoperability.

The **integrability** attribute is addressed by the usage of the Façade pattern and Abstract Data Repository style.

- In the facade of a component the methods and parameters are grouped together and offered for access hiding internal component implementation. An agreement on a façade allows separate development of components and their easier integration.
- The usage of the Abstract Data Repository style allows data producers and consumers to exchange data unaware of each others requirements on the data format. Consequently, abstract data repositories become “points of data integration”.

The **security** attribute is addressed by the introduction of the Secure Messenger and the optional Signer components. Manipulation and storage of sensitive business data (contracts and contracting data) is delegated to dedicated components, i.e., Updater and Contract distributor (MC/SA).

The **flexibility** attribute is addressed by the introduction of an Enactment Server component that supports the flexible execution of the contracting process. ERA-based systems can establish relationships with diverse parties and can make use of diverse mediators.

Based on the analysis of the relations between architectural decisions and quality attributes, we have defined the following sensitivity and tradeoff points and the risks and non-risks in ERA:

Sensitivity points

- S1: Mappers can map data and processes between standards that have a common conceptual foundation. It may not be possible to absorb in a mapper external messages/protocols that differ conceptually from the internally used standards. The limitations of mappers can decrease the interoperability attribute of ERA.
- S2: The complexity of the ES, Design, Negotiator, and EES components may hinder development of an ERA-based system (and thus, affect the buildability and automation qualities of ERA).

Tradeoff points

- T1: It provides security, but affects performance.
- T2: Mappers will require substantial computational power and may cause decrease in performance.
- T3: The choice of the Layering style for the structure of ERA affects performance as it requires communication to take place only between neighboring layers.
- T4: The usage of the Abstract Data Repository style can affect performance.

Risks

- R1: The secure messenger is a potential bottleneck. This problem must be addressed in another architectural view.
- R2: The mapping components may be insufficient to incorporate substantial differences which will lead to changes in internal components.
- R3: The highly advanced functionalities of certain components may be difficult to implement.

Nonrisks

- N1: The load on the Signature Manager is decreased which leads to improved performance.
- N2: As there are only 3 layers performance should not be seriously affected.
- N3: Part-whole decomposition has no negative effects on other quality attributes [7].

3.2. Testing of ERA via ATAM

The next steps in ATAM are related to testing the architecture after its evolution in the analysis phase. As a first step, a meeting with the stakeholders of the architecture has to be conducted. ATAM is used for evaluation of concrete architectures that have a clearly defined group of stakeholders. However, in the case of a reference architecture, there is no concrete group of stakeholders. We identified as main stakeholders a number of roles, i.e., contract managers (this role is given different names like contract engineers, contract supervisors, contract legal officers, etc.), software architects/designers, CIOs, and CEOs. To approach some of the potential stakeholders, we organized a workshop on e-contracting [13], where we presented ERA to 25 experts working as contract managers, CIOs, or CEOs. Though this meeting cannot be representative for all stakeholders of ERA, we consider it sufficient to indicate the level of excellence of ERA (addressing all stakeholders would indeed be impossible). After an introduction to electronic contracting, we asked participants to provide their expectations in terms of quality attributes from ERA. The resulting utility tree was a subset of the attributes identified initially by us (see Figure 4). Two of the qualities defined by the participants were not applicable to the design of ERA. "Portability" is not of importance for ERA, as ERA is technology independent. Certification is beyond our research goals.

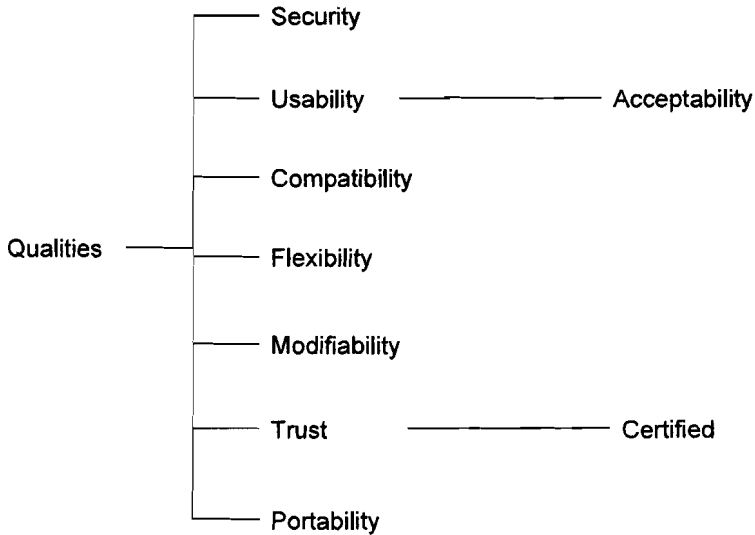


Figure 4 - Utility tree defined by the workshop participants

We asked the participants in the workshop to suggest possible scenarios and prioritize them. The result was a set of 25 use-case scenarios (see Appendix B). After the workshop, we analyzed these scenarios. Based on the analysis, we made a final adaptation of ERA. For our surprise, the most important scenario defined by the participants was not addressed properly in ERA. In our post-workshop analysis, we realized that the reason for omission of this activity was the lack of thorough investigation of the internal activities performed by the contracting parties prior to the contracting process. Thus, the main outcome from this step was the introduction of a new functionality to ERA, i.e., the introduction of the “Knowledge Updater” component.

Though the workshop proved useful for the evaluation of the functionalities of ERA, it revealed a number of problems of using ATAM for evaluation of reference architectures. First, ERA is a reference architecture defined for a highly advanced information system that is currently addressed only in the research world. However, the stakeholders of ERA were mostly interested in practical solutions that can be directly implemented. Though, we spent a significant amount of time in presenting the goals of ERA, as it can be seen from the scenarios suggested by the workshop participants, their expectations were still for a system with low-level of automation. We had to use the scenarios generated by the participants and re-state them for the advanced situation. Second, some stakeholders did not see the value of a reference e-contracting architecture. This, combined with their lower expectations from an e-contracting system, resulted in lack of sufficient involvement of the workshop participants in the discussions on the final version of ERA.

3.3. Evaluation of architectural non-functional requirements

ATAM is not designed for the evaluation of architectural non-functional requirements. Our attempt to use ATAM for the evaluation of the architectural non-functional requirements failed as early as we attempted to define concrete scenarios for them in the utility tree. That is why we used different techniques for the evaluation of these qualities in ERA. The meetings and the workshop that were part of the ATAM process proved to be useful for the evaluation of the usability requirement. We monitored the level of understanding from the diverse stakeholders that participated in these meetings. Our conclusion is that ERA can be easily understood by both business persons as well as architecture designers.

For the evaluation of the applicability quality of ERA, we applied ERA for analysis of a number of e-contracting architectures defined both in academia and industry. The comparisons can be found in [3]. Our conclusion after the performance of this step was that ERA is more complete than existing e-contracting architectures. Furthermore, the abstraction and decomposition techniques that we used in its design allowed us to successfully apply it for analysis of architectures with different levels of detail.

We evaluated the functional qualities of ERA by means of research results stating functional requirements on an e-contracting system, comparison with existing systems, and by obtaining feedback on its functionalities during the workshop that we organized. Based on these steps, we concluded that ERA in its final version addresses the complete set of functionalities required in an e-contracting system.

For the evaluation of ERA for its level of automation, we reasoned on the automation provided by the components defined in the final version of the architecture. We concluded that only activities required for the definition of the specific business context may have to be performed manually (definition of supported activities, of applicable business rules, of contract templates, negotiation strategies, etc.).

The structure and conceptual integrity of ERA (part of the feasibility quality) are demonstrated by the list of architectural approaches that we elaborated in ATAM and the reasoning on the usage of this approaches that we present in this report. For the buildability quality of ERA (also part of its feasibility), we suggest readers to refer to [14] for further information.

4. Conclusions

In this report, we presented details on the process of evaluation of the initial design of an E-contracting Reference Architecture presented in [2]. The results from this evaluation were used to define an improved version of ERA [3].

For the evaluation process, we used in its major part the Architecture Tradeoff Analysis Method (ATAM), which is a widely accepted method for evaluation of software architectures. However, as this method is not targeted to the evaluation of reference architectures, we had to extend and tailor it for the evaluation of ERA.

The evaluation process has allowed us to achieve a substantial evolution of ERA from its initial design. ERA improved significantly in its structure and conceptual integrity. The application of ATAM forced us to apply established approaches for data exchange and component invocation. The functional completeness of ERA was also improved. Based on the evaluation process, we can conclude that improved version of ERA excels in both system and architecture non-functional requirements.

Acknowledgments

We thank Paul Grefen and Jochem Vonk, who helped us in the evaluation of the architectural approaches used in ERA. We thank all workshop participants for their contributions for the improvement of ERA.

Appendix A Analysis of the Architectural Approaches

Analysis of Architectural Approach				
Scenario: 1. Only original recipients of messages can read them (counterparties/mediators).				
Attribute(s) External environment				
Environment Normal environment				
Stimulus The system sends a message to a counterparty/mediator				
Response The message is read only by the recipient				
Architectural decisions	Sensitivity	Tradeoff	Risk	Nonrisk
<i>Cryptographer component</i>	-	T1	R1	-
Reasoning The cryptographer component is defined to deliver this functionality.				

Analysis of Architectural Approach				
Scenario: 2. External messages cannot be tampered				
Attribute(s) External environment				
Environment Normal				
Stimulus The system sends a message				
Response The message is received by the recipient unchanged or the change is detected.				
Architectural decisions	Sensitivity	Tradeoff	Risk	Nonrisk
<i>Signature manager component</i>	-	T1	R1	-

Reasoning

The Signature Manager guarantees to the receiver identification of any changes on the message.

Analysis of Architectural Approach

Scenario: 3. Parties use e-notaries, cert. auth., etc. to increase trust in e-contracting

Attribute(s) Trust

Environment Normal environment

Stimulus A party decides to use a "trust-mediator" for the contracting process

Response The mediator's functionalities can be used by the system

Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
The use of a mediator is allocated to specific components - Certificate authority - in the Signature Manager - E-notary - in the Contract Establisher - Arbitrator - Dispute Resolution Manager <i>Part-whole decomposition</i>	-	-	-	-

Reasoning

A sub-component knows how to use a mediator. The use of additional mediators will require system modifications (introduction of new functionality) (see modifiability)

Analysis of Architectural Approach

Scenario: 4. The actor signing a contract is known

Attribute(s) Internal

Environment Normal environment

Stimulus Contract is signed

Response	Parties can clearly identify the actor that signed the contract			
Architectural decisions	Sensitivity	Tradeoff	Risk	Nonrisk
<i>Signer</i> components	-	-	-	N1

Reasoning

The Signer can manage authorities/responsibilities in the signing of a contract. The Signer takes part of the load of the Signature Manager.

Analysis of Architectural Approach				
Scenario:	5. No conflicts may arise from creation/updating of contracting data.			
Attribute(s)	Trust			
Environment	Normal			
Stimulus	A component creates/updates contracting data			
Response	There is no conflict with any other component on the data creation/update.			
Architectural decisions	Sens,	Tradeoff	Risk	Nonrisk
<i>Updater</i> component <i>Contract distributor (MC/SA)</i> component <i>Data Manager</i> component	-	-	-	-

Reasoning

By designing a single component that can manipulate contracting data, direct conflicts are avoided. When the same data can be created/updated by a number of components, we define a dedicated component to control the data creation/update.

Analysis of Architectural Approach	
Scenario:	6. No conflicts arise from creation/updating of business rules

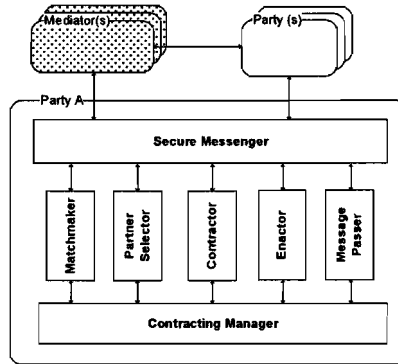
Attribute(s) Trust				
Environment				
Stimulus	The company wants to introduce a new business rule to the system			
Response	The business rule introduced to the system is not in conflict with existing business rules			
Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
<i>Separation of concerns (part-whole decomposition, layering style)</i> <i>Abstract data style</i>		T4		
Reasoning By clearly defining where different types of business rules are stored, we separate concerns and eliminate overlaps among storages of business rules and thus of conflicts among them.				

Analysis of Architectural Approach				
Scenario:	7. An ERA-based system is extended with new functionalities			
Attribute(s)	Modifiability			
Environment -				
Stimulus	The company wants to introduce a new functionality in the system			
Response architecture	The functionality is introduced without major changes to the architecture			
Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
Layering pattern	-	T3	-	N2
Part-whole decomposition				N3
Data indirection pattern				
Facade pattern				
Reasoning				
- Layering introduces separation of concerns. Changes in functionalities influence only neighboring layers. - Part-whole decomposition promotes system modifiability (separation of concerns). - We expect that a second-level component will be developed by a single company. Consequently, differences in data format between data providers and data consumers will be less problematic. For this reason, for data exchange within components, we opted for the Data				

indirection pattern, which supports easy modifications (additions/removal of data consumers/providers). If differences in the data format are expected it can be opted for the Abstract data repository pattern.

- Facade pattern facilitates component encapsulation.

Architectural diagram



Analysis of Architectural Approach

Scenario: 8. Change of trading standards (protocols, message templates, legal framework)

Attribute(s) Modifiability

Environment Normal

Stimulus A new "standard" is introduced (for workflow process specification, communication protocol, message/document semantics/representation, etc.).

Response The system is adapted with minor efforts, without changes on its core components

Architectural decisions	Sens.	Trade off	Risk	Nonrisk
External Mapper component	S1	T2	R2	
Internal Mapper component				
Enactment Mapper component				

Reasoning

- Changes in external message standards will be reflected only in the External Mapper.
 - Changes in internal messages for invocation of the Contract Manager are reflected only in the Internal Mapper
 - Changes in the internal process specifications require adaptations only in the Enactment Mapper.
-

Analysis of Architectural Approach

Scenario: 9. An ERA-based system sends (or receives) standard-conforming messages that can be automatically interpreted.

Attribute(s) Interoperability

Environment Normal

Stimulus The system sends or receives a standardized message

Response The message is interpreted by the party/counterparty

Architectural decisions	Sensitivity	Tradeoff	Risk	Nonrisk
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<i>External Mapper</i> component

S1

R2

Reasoning

The mapping component allows achieving integration between data definitions in the e-contracting system and data definitions in external systems

Analysis of Architectural Approach

Scenario: 10. An ERA-based system sends (or receives) data messages to (from) internal systems that can be automatically interpret.

Attribute(s) Interoperability

Environment -

Stimulus An ERA-based system sends (or receives) data messages to (from) internal systems

Response The data can be interpreted by the receiving side

Architectural decisions	Sens.	Trade off	Risk	Nonrisk
<i>Internal Mapper</i> component <i>Enactment Mapper</i> component	S1		R2	

Reasoning

The mapping components allow to achieve integration between data definitions in the e-contracting system and data definitions in other internal systems.

Analysis of Architectural Approach

Scenario: 11. An ERA-based system can support diverse e-contracting processes that fit best the business context

Attribute(s) Flexibility

Environment Normal

Stimulus A counterparty requires the execution/skipping of certain activities

Response The system adapts to the partner process requirements

Architectural decisions	Sens.	Trade off	Risk	Nonrisk
<i>Communication Monitor</i> component <i>Enactment Server</i> component <i>Matchmaker</i> component <i>External Mapper</i> component	S2		R3	

Reasoning

The Matchmaker component returns after its execution matching parties (including the supported processes by counterparties). The Enactment Server uses this information when invoking other components – it instructs the invoked components which activities they are allowed to perform and under which conditions. The Communication Monitor monitors if the components and the counterparties stick to the agreed rules. The External Mapper maps the internal process specifications to external, common process specification standards.

Analysis of Architectural Approach				
Scenario:	12. An ERA-based system exerts control over the enactment process at the counterparty (as agreed in the contract)			
Attribute(s)	Interoperability			
Environment	Normal			
Stimulus	The system decides to make a change (agreed in the contract) at the execution of the agreed enactment process			
Response and	The system communicates the process change to the counterparty the party interprets it and complies with it.			
Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
<i>External Enactment Server</i> component				
<i>Data Manager</i> component				
Reasoning The EES makes decisions on the contract enactment plan. When it evaluates the need for a control, it communicates this to the Data Manager that informs the counter party for the required change. The Data Manager verifies any requests from counterparties on process control and if approved informs the EES for the required change.				

Analysis of Architectural Approach				
Scenario:	13. Messages (events and data) from contract agreed processes are communicated to and interpreted by the relevant internal systems			
Attribute(s)	Interoperability			
Environment	Normal			
Stimulus	The system informs an internal system for a contract enactment event that is relevant for it			
Response	The internal system receives and interprets the event.			
Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
<i>Enactment Mapper</i> component				
Reasoning The Enactment Mapper maps both data and event information.				

Analysis of Architectural Approach				
Scenario: 14. Components of an ERA-based system can invoke each others functionalities without specific tailoring of the invocation format.				
Attribute(s) Integrability				
Environment -				
Stimulus A component sends message to another component				
Response The receiver can interpret the message				
Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
<i>Facade pattern</i>				
Reasoning The Facade pattern isolates components from one layer from components belonging to the upper layer. In the facade of each component the methods and parameters are grouped together to the upper layer. These can differ from internal for the component implementation. A standard for the facades allows the development of components separately and their easy integration.				

Analysis of Architectural Approach				
Scenario: 15. Components of an ERA-based system that are developed independently can exchange data without specific component tailoring.				
Attribute(s) Integrability				
Environment -				
Stimulus Component sends data to another component				
Response The receiving component can interpret the data sent to it without complex component tailoring				
Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
<i>Abstract data repository pattern</i>				
Reasoning The Abstract data repository pattern is used to isolate data send from a component to another component. The format of the data provider does not have to match the format of the data consumer that belongs to another layer.				

Analysis of Architectural Approach

Scenario: 16. An ERA-based system can support e-contracting with diverse Partners.

Attribute(s) Flexibility

Environment -

Stimulus A company wants to start contracting with a new partner.

Response The system adapts to the new situation and can support the whole process with the new potential partner.

Architectural decisions	Sens.	Tradeoff	Risk	Nonrisk
<i>Knowledge updater component</i>				

Reasoning

The Knowledge Updater component obtains process and contract ontologies of other parties and mappings to the ontology supported by the company. This allows the company to interact with diverse new partners.

Sensitivity points

- S1: Mappers can map data and processes between standards that have a common conceptual foundation. It may not be possible to absorb in a mapper external messages/protocols that differ conceptually from the internally used standards. The limitations of mappers can decrease the interoperability attribute of ERA.
- S2: The complexity of the ES, Design, Negotiator, and EES components may hinder development of an ERA-based system (and thus, affect the buildability and automation qualities of ERA).

Tradeoff points

- T1: It provides security, but affects performance.
- T2: Mappers will require substantial computational power and may cause decrease in performance.
- T3: The choice of the Layering style for the structure of ERA affects performance as it requires communication to take place only between neighboring layers.
- T4: The usage of the Abstract Data Repository style can affect performance.

Risks

- R1: The secure messenger is a potential bottleneck. This problem must be addressed in another architectural view.
- R2: The mapping components may be insufficient to incorporate substantial differences which will lead to changes in internal components.
- R3: The highly advanced functionalities of certain components may be difficult to implement.

Nonrisks

- N1: The load on the Signature Manager is decreased which leads to improved performance.
- N2: As there are only 3 layers performance should not be seriously affected.
- N3: Part-whole decomposition has no negative effect on other attributes [7].

Appendix B

Scenarios defined by the workshop participants

1. Adding contractors	
2. Comparison between reality and planned in contract schedule.	Votes: 1
3. Ability to interpret clauses in a contract	Votes: 4
4. Can easily interface with other software	Votes: 4
5. Having the most recent and approved contract templates and terms and conditions in the system	Votes: 6
6. List activities that are expected to be completed in the next month (automatic notification)	Votes: 4
7. Get alerts for unresolved claims	Votes: 1
8. Provide information on the maximum liability	Votes: 1
9. Integrated information delivery.	
10. Advisory function on contract creation	
11. Accessible from everywhere	Votes: 3
12. Multilingual support	Votes: 2
13. Reporting	Votes: 5
14. Large storage capacity	Votes: 1
15. Easy data search	Votes: 3
16. Help	
17. Document enabled	
18. Currency support (exchange rates, etc.)	Votes: 1
19. Voice support	
20. Dictionary	Votes: 1
21. Easy to upgrade	
22. Renewals, amendments, licensing	Votes: 1

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